MA 102 (Mathematics II - Ordinary Differential Equations)

Department of Mathematics, IIT Guwahati
Semester: January-April, 2024

Tutorial Sheet No. 4   Date of Discussion: April 12, 2024

Lectures 10-14: Singular points, Power series solution for second-order ODE, Bessel functions, Legendre polynomials

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Attention students: Please keep in mind that it is not possible to discuss all problems during the tutorial. You can consider the problems not discussed as your practice problems.

1. Determine the radii of convergence of the following power series:

   (a) $\displaystyle\sum_{n=0}^{\infty} \frac{n^2}{2^n}(x+2)^n$; (b) $\displaystyle\sum_{n=1}^{\infty} \frac{3}{n^3}(x-2)^n$; (c) $\displaystyle\sum_{n=0}^{\infty} \frac{2^{-n}}{n+1}(x-1)^n$.

2. Classify the singular points of the following differential equations:

   (a) $e^x y'' - (x^2-1)y' + 2xy = 0$; (b) $(x^2+x)y'' + 3y' - 6xy = 0$;
   (c) $(\sin x)y'' + (\cos x)y = 0$; (d) $\ln(x-1)y'' + (\sin 2x)y' - e^x y = 0$, for $x > 1$.

3. Find the indicial equation and hence their roots of the following differential equations:

   (a) $(x^2-x-2)y'' + (x^2-4)y' - 6xy = 0$; (b) $x^2 y'' + 2(x-3x^2)y' + e^x y = 0$;
   (c) $x^2 y'' + (\sin x)y' + (\cos x)y = 0$.

4. Find a series solution about the ordinary point $x = 0$ of each of the following differential equations subject to the given initial conditions:

   (a) $(4-x^2)y'' + 2y = 0$, $y(0) = 0$, $y'(0) = 1$;
   (b) $y'' - x^2 y' + y\sin x = 0$, $y(0) = 0$, $y'(0) = 1$.

5. Find a series solution about the regular singular point of each of the following differential equations:

   (a) $xy'' + 4y' - xy = 0$; (b) $4x^2 y'' + 2x^2 y' - (x+3)y = 0$.

6. Use the orthogonality property of Legendre polynomials to show that
   $\displaystyle\int_{-1}^{1} g(x)P_n(x)dx = 0$ for every polynomial $g(x)$ with $\deg(g(x)) < n$.

7. Find the specific solution to the differential equation $(1-x^2)y''-2xy'+12y=0$ subject to $y'(0)=4$ and the function $y(x)$ is well-behaved at $x=1$.

8. Using the substitution $\cos\theta = x$, show that the differential equation

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{dy}{d\theta}\right)+n(n+1)y=0$$

can be converted to the standard form of Legendre equation.

9. Find a general solution to the following differential equations in terms of Bessel functions:

(a) $4x^2y''+4xy'+(4x^2-1)y=0$; (b) $x^2y''+xy'+x^2y=0$;
(c) $x^2y''+xy'+(16x^2-1/9)y=0$; (d) $x^2y''+xy'+((4/9)x^2-9/4)y=0$.

10. The differential equation $x^2y''+(x^2-2)y=0,\ x>0$ does not look like a Bessel's equation. However, show that, by making a substitution $y=v(x)\sqrt{x}$ of the dependent variable, its solution can be expressed in terms of Bessel functions.

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